

.NET JavaScriptSerializer Deserialization Vulnerability

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Content (English translation)

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[()]Introduction

Original text from the Microsoft home page:

For .NET Framework 4.7.2 and later versions, use the APIs in the System.Text.Json namespace for serialization and deserialization. For earlier versions of .NET Framework, use Newtonsoft.Json. This type was intended to provide serialization and deserialization functionality for AJAX-enabled applications.

They provide Serialize and deserialize features in .NET for the json format, and along with them there are some supporting libraries such as: [Json.NET](#), [DataContractJsonSerializer](#) and [JavaScriptSerializer](#)

[Json.NET](#) (Newtonsoft.Json): This is an open source library developed by James Newton-King, not owned by Microsoft. However, [Json.NET](#) is very widely used in .NET projects, and Microsoft integrated [Json.NET](#) into [ASP.NET](#) Core as the standard solution for handling JSON in some versions, before moving to System.Text.Json.

[DataContractJsonSerializer](#): This is a part of the .NET Framework. The library is developed and maintained by Microsoft.

JavaScriptSerializer: This is also a Microsoft library. It lives in the System.Web.Script.Serialization namespace and is mainly used to convert data between JSON and objects in [ASP.NET](#) applications.

[(())JavaScriptSerializer

[(())Demo:

- I use JetBrains Rider to create the project and debug

Then add `System.Web.Extensions` in Reference under Dependencies

[(())Code:

- The code above shows 2 ways of serializing, with and without SimpleTypeResolver.
- Output:

[(())JavaScriptSerializer

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At line 23 it creates `JavaScriptSerializer` with the arg `type resolver`

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SimpleTypeResolver is a class inherited from JavaScriptTypeResolver, used for deserializing and serializing the typed json form

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Going deeper into the JavaScriptSerializer class, we see there are 3 methods for deserialization:

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All 3 of these methods return: `return JavaScriptSerializer.Deserialize(this, input, (Type) null, this.RecursionLimit);`

To go deeper into the functions inside, I use [ysoserial.net](#) to build the chain and then analyse it.

Then add the payload into the json file in the following folder:

[(())Chain analysis:

Code:

Starting from serializerWithType.Deserialize it calls the Deserialize function

-> BasicDeserialize

-> DeserializeInternal Line 67: converts the data into a dictionary made of its keys and values

Line 68 checks whether DeserializeDictionary contains the key `__type` or not. In this case it does, so it goes on to call ConvertObjectToType

-> ConvertObjectToTypeMain

Line 260 checks the type for null -> calls ConvertObjectToTypeInternal

-> Calls ConvertDictionaryToObject

-> serializer.TypeResolver.ResolveType(id) -> dictionary.Remove("__type") -> Activator.CreateInstance(type1)

When it goes into `__type`, the json holding the fields is converted into an object of the matching type.

[]()CVE-2019-18935

An overview of this CVE:

When Telerik uploads, it produces `rauPostData` (roughly, an encrypted blob made of a few parts) which is then used to check whether anything was modified during the upload.

For example in this piece of code:

Combining CVE-2017-11317 (arbitrary file upload) + the Deserialize bug creates this CVE

[]()The vulnerable function:

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Line 373 splits `rauPostData` into 2 array elements using split `&`; the first part is assigned to the variable text, and the remaining part is the type.

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Line 376 deserializes with the specified type

Going deeper into the SerializationService class we see it uses JavaScriptSerializer; line 32 invokes the kind of object we pass in, and here we control it into json form so we can get RCE.

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At the BlackHat event they published a Json Deser gadget: <https://www.blackhat.com/docs/us-17/thursday/us-17-Munoz-Friday-The-13th-JSON-Attacks-wp.pdf>

They also made clear the 2 directions for getting RCE, as in the picture. So in this CVE we can take advantage of the arbitrary file upload: I will upload a Mixed Assembly dll file. Then I will use the chain to call that dll file to trigger it.

[]()POC:

File calc.c:

Then compile the c code above into an assembly DLL. Combine and adapt the file upload CVE POC: https://github.com/bao7uo/RAU_crypto/blob/master/RAU_crypto.py

After running the request Calc pops up, so it worked.

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